

Aayush Bathija

bathija.aayush@gmail.com

EDUCATION

Oak Park High School Class of 2029	Oak Park, CA
Moorpark College, dual enrollment Calculus I, II. Psychology, Symbolic Logic, Philosophy, and General Physics I	2022–present
Independent study. Real analysis through Tao’s <i>Analysis I</i> .	

PAPERS

- A. Bathija, P. Rohatgi, D. Soskin. *Bounded Ratios for Lorentzian Polynomials*. arXiv:2609.05341, 2026.

RESEARCH

Bounded ratios for Lorentzian polynomials Extending the Huang–Huh–Soskin–Wang results to general degree: a dual description of the bounded-ratio cone in any degree, optimal constants for ternary Lorentzian cubics, and a classification of when the Lorentzian and linear forms Polynomial Coincide. With Prince Rohatgi and Prof. Daniel Soskin (UCLA). Funded by ORMC’s Vertical Research Integration program.	2026–present
Ahlsvede–Daykin for determinantal inequalities Applying the Ahlsvede–Daykin four functions theorem to products of minors of totally positive matrices, yielding interval inequalities on the sorting lattice. Manuscript in preparation with Prince Rohatgi and Prof. Daniel Soskin (UCLA). Funded by ORMC’s Vertical Research Integration program.	2026–present
Fairness of voting systems Simulating ranked-choice, STAR, and related methods and comparing how each one fails. Started in the Non-Trivial fellowship and ongoing.	2026–present

TEACHING

Instructor, UCLA Olga Radko Math Circle Teaching weekly sessions, and taking pure math classes.	2026–present
Instructor, Agoura Math Circle Taught and graded weekly classes. Primary on the documentation team, and wrote scripts that migrate documentation.	2024–2026
Volunteer coach, Medea Creek Middle School Running practice for the school’s MATHCOUNTS team.	2025–present

LEADERSHIP

- President, Philosophy Club, Oak Park High School. Preparing presentations, organizing talks, and facilitating discussion.
- Founder, Higher Math Club, Oak Park High School. Proof-based material for students who have exhausted competition math.

AWARDS

- Guinness World Record, most countries identified by outline in one minute
- John Locke Institute Essay Competition, 2x shortlist
- Superintendent’s Compassionate and Creative Citizens Award, given to four students district-wide and chosen by the superintendent
- Great History Challenge, national runner-up
- Berkeley Math Tournament, top ten in the Power Round (team)
- AMC 10, Distinction in 2026 (AIME qualifier)

PROGRAMS

AwesomeMath Academy Number Theory with Mirroslav Yotov, into algebraic number theory: Gaussian integers, unique factorization, quadratic residues. Also Computational Geometry and Counting Strategies.	2024–present
Non-Trivial Research Fellowship Voting systems project	2025
LEAF Program, <i>The Mathematics of Morality</i>	2024

OTHER

Academic Learning (youtube.com/@Academic-Learning), discussing and working through mathematical material. Speech and debate: reached the final round of the state qualifier. Speedcubing: 6.58 s single, 3×3 .